

ANSWERS

1.

Question	Answer	Marks
9(a)	any two from: from <u>gravitational</u> potential energy or <u>gravitational</u> potential energy <u>to</u> or <u>gravitational</u> potential energy decreases kinetic energy mentioned <u>to</u> internal energy or internal energy increases	B2
9(b)(i)	different number of neutrons (in the nucleus)	B1
9(b)(ii)	(its nucleus only contains) only has one proton (alpha-particles contain two protons) or not enough protons	B1
9(b)(iii)	${}^0_{-1}(\beta)$	B1
	${}^3_2(\text{He})$	B1
9(b)(iv)	(nuclear reaction produces) a high temperature (in core)	B1
	outward force (due to high temperature / nuclear reaction) or force due to high temperature / nuclear reaction or radiation pressure	B1
	balances (gravitational) force (inwards)	B1
9(c)	(red supergiant explodes as) supernova (explosion)	B1

2.

Question	Answer	Marks
10(a)(i)	horizontal path marked with γ and no other path indicated as γ	B1
10(a)(ii)	beta (β -radiation)	B1
	left-hand rule (mentioned)	B1
	first finger into page and thumb downwards	B1
	current opposite to motion or particles negative (and so are beta)	B1
10(a)(iii)	beta (β -radiation) completely absorbed	B1
	<u>some</u> gamma rays absorbed or (some) gamma passes through	B1
10(b)	$(\lambda =) v/f = 3.0 \times 10^8 / 8.8 \times 10^{19}$	C1
	$3.4 \times 10^{-12} \text{ (m)}$	A1

3.

Question	Answer	Marks
8(a)	it / uranium-235 has (3) fewer neutrons (in the nucleus)	B1
8(b)(i)	the nucleus splits (and releases more neutrons)	B1
8(b)(ii)	neutrons released (when nucleus splits)	B1
	neutrons absorbed by other (uranium-235) nuclei	B1
	these nuclei split and release neutrons	B1
	this process continues (using released neutrons)	B1
8(b)(iii)	to slow down the neutrons (emitted by fission)	B1
8(b)(iv)	they / control rods absorb neutrons (without further fission)	B1
	to slow down / stop / control the chain reaction / limit the number of fissions	B1

4.

Question	Answer	Marks
9(a)(i)	alpha	B1
9(a)(ii)	beta	B1
9(b)(i)	beta particles because count rate goes down (by 600) when metal placed in position	B1
9(b)(i)	beta absorbed by metal / cannot pass through metal	B1
	gamma (because) radiation passes through metal / still gives reading with thick metal	B1
9(b)(ii)	alpha stopped by a few cm in air / alpha do not reach G.M. tube or alpha not detected by G.M. tube (end cap stops them)	B1
9(c)	use tongs / handle at distance / place absorber between source and observer / limit exposure time / place in lead box or thick metal box	B1

5.

Question	Answer	Marks
10(a)	emission from the <u>nucleus</u>	B1
	of an alpha particle or beta particle or gamma (radiation)	B1
10(b)	at least three points plotted at (6000, 800), (12 000, 400), (18 000, 200) and (24 000, 100)	C1
	all points correct and smooth curve drawn	A1
10(c)	GM (tube and) counter (and timer)	B1
	measure number of counts in a specified time (e.g. 1 min) or subtract background count or tube connected to ratemeter	B1
10(d)	measure count rate of similar (mass) sample of living / fresh wood	B1
	use the graph to find time or description of use of half-life to find age	B1

6.

Question	Answer				Marks
9(a)		yes	no	it is not possible to tell	B1
	it emits an α -particle		✓		
	it emits a β -particle	✓			
		yes	no	it is not possible to tell	B1
	it emits a γ -ray			✓	
9(b)(i)	(both have) 91 protons / equal numbers of protons (in nucleus)				B1
9(b)(ii)	(they have) different numbers of neutrons (in nucleus) / nucleons				B1

9(c)(i)	(count that is) always present or due to the environment / surroundings or not deliberately introduced or not being tested	B1
	any two from: rock sources (e.g. benches, building materials, human body, Earth crust) air sources (e.g. radon) cosmic sources (e.g. Sun, space)	B2
9(c)(ii)	use tongs / tweezers / forceps or wear lead(-lined) clothing / lead(-lined) gloves or wear goggles or reduce time of exposure or keep distance of separation large	B1
9(c)(iii)	(radioactive) emission is a random process / occurs at random intervals	B1
	measurements fluctuate (about an average value)	B1
9(c)(iv)	(coordinates of) two points from curve, e.g. (0, 58) and (70, 29)	C1
	(coordinates of) two points from curve and at least two half-lives apart, e.g. (0, 58) and (138, 14.5) or $64.0 \leq \tau \leq 74.0$	C1
	$66.0 \text{ s} \leq \tau \leq 73.5 \text{ s}$	A1
9(c)(v)	(large half-life because of) slow rate of decay	B1
	radiation / count rate due to $^{234}_{92}\text{U}$ imperceptible / very much less than that of $^{234}_{91}\text{Pa}$ or (nearly) all of measured radiation / count rate due to $^{234}_{91}\text{Pa}$	B1

7.

Question	Answer	Marks
9(a)(i)	helium atom has (any 1 from) - two electrons (and ion has 1 electron) - one more electron (than the ion) - no charge (and ion positive) - same number of electrons as protons;	B1
9(a)(ii)	alpha particle has (any 1 from) : - no electrons - charge +2 (e) - only protons and neutrons or He ion has an electron	B1
9(a)(iii)	<i>X number of protons</i> 88	B1
	<i>Y number of protons</i> 89 or number of protons for X +1	B1
	<i>X number of neutrons</i> 137 or 225 – number of protons in X	B1
	<i>Y number of neutrons</i> 136 or number of neutrons for X –1	B1
9(b)	GM tube	cloud chamber
	use paper absorber and / or (thin) metal / >5 cm air or deflect with magnetic / electric field / charged plates	look at trails in cloud chamber <u>when</u> vapour condenses / cools
	with paper, reading / count drops if alpha present or alpha particles stopped by paper / >5 cm air or correct deflection and sensible detection of alpha particles in field	alpha tracks straight / thick / few cm long
	with metal, reading / count drops (further) if beta present or beta particles stopped by metal or correct deflection and sensible deflection of beta particles in field	beta tracks spidery / thin / long
9(c)(i)	time taken to halve	C1
	time for number of (radioactive) atoms / nuclei / count / activity to halve	A1
9(c)(ii)	any halving of 4.0×10^{14} seen	C1
	only 3 halvings or 3 half lives seen	C1
	7300 years	A1

8.

Question	Answer	Marks
9(b)(i)	technetium(–99)	B1
9(b)(ii)	only <u>gamma</u> detected outside body	B1
	EITHER half-life / 6 hours is short and less damage (to others / long term) / loses radioactivity quickly OR half-life / 6 hours is long enough for isotope to reach site and be detected / long enough to examine patient / does not need to be replaced	B1
9(c)(i)	cosmic rays; rocks; buildings; food and drink etc.	B1
9(c)(ii)	weapons tests; nuclear weapons; nuclear power stations; nuclear accidents, e.g. Chernobyl;	B1
9(c)(iii)	cancer / kill cells / harm DNA / RNA / cause mutations	B1
9(d)(i)	superscript 1 and subscript 0	B1
9(d)(ii)	neutron	B1
9(d)(iii)	in the Sun / star or at high temperature	B1
9(e)	alpha-particles more easily stopped (than beta)	B1
	alpha-particles cause more ionisation (than beta)	B1
	deflected in opposite directions (by magnetic field) or alpha-particles deflected less (for same speed)	B1

9.

Question	Answer	Marks
6(a)	any two from: number of protons decreases by two number of neutrons decreases by two or number of nucleons decreases by four	B2
6(b)(i)	background radiation (detected) or source of background radiation mentioned (e.g. cosmic rays; radiation in the air)	B1
6(b)(ii)	alpha-particles are absorbed by the <u>air</u> (between the source and detector)	B1
	all the alpha-particles / radiation are absorbed (by the air between the source and detector) or alpha-particles travel less than 10 cm (in air)	B1
6(c)(i)	arrow through J towards centre of circular path	B1
6(c)(ii)	into the page (i.e. fifth box ticked)	B1
6(c)(iii)	force perpendicular to direction of motion or no component of velocity along line of action of force or work done = force × distance moved in direction of motion	M1
	no work done	A1

10.

Question	Answer	Marks
10(a)	same number of protons (in nucleus) or same proton number or same number of electrons in a neutral atom or same atomic number	B1

10(b)	one more proton	B1
	one fewer neutron	B1
	one more electron	B1
10(c)(i)	18 counts / minute	B1
10(c)(ii)	any two of: rocks / building (materials) / coal / radon space / Sun / cosmic rays weapons testing / nuclear waste	B2
10(c)(iii)	one count rate – 18 or one count rate – 18 calculated e.g. 74 – 18 or 56	B1
	one count rate – 18 halved or one count rate – 18 reduced by factor 2^N e.g. 56 / 2 or 28 or 56 / 4 or 14	B1
	one count rate – 18 reduced by factor $2^N + 18$ e.g. 28 + 18 or 46 or 14 + 18 or 32	B1
	65 hours	B1
10(c)(iv)	(radioactive decay is) random	B1
	in time or impossible to predict when a beta-particle is emitted	B1
10(d)	correct upward curved path between plates	M1
	no more than 1 cm straight at beginning	A1

11.

Question	Answer	Marks
11(a)	electron	B1
11(b)	prevents exposure to radiation / for safety (of experimenter and others)	B1
	beta particles stopped / absorbed by metal / case or prevents radiation travelling in all directions / to surroundings	B1
11(c)(i)	decreases (slightly) / little change	B1
	particles spread out or <u>some</u> stopped / absorbed (by air)	B1
11(c)(ii)	no particles detected or <u>large</u> decrease	B1
	(>1m) <u>air</u> stops (beta) particles / <u>air</u> atoms / particles become ionised	B1
11(d)	proton number of Y = 39	B1
	'nucleon' number of $\beta = 0$	B1
	nucleon numbers of Y and β add to 90	B1
11(e)(i)	<i>in (nuclear) fusion</i> <u>nuclei</u> come together	B1
	<i>in (nuclear) fission</i> a <u>nucleus</u> splits	B1
	either in (nuclear) fusion <u>small</u> nuclei come together or in (nuclear) fission <u>large</u> nuclei split or fission caused by neutron (hitting nucleus)	B1
11(e)(ii)	high temperature or particles moving at high speed	B1
	high pressure or high density (of particles)	B1

12.

Question	Answer	Marks
10(a)	<i>alpha</i> positive (+2)	B1
	<i>alpha</i> stopped by paper or thin metal or 3–10 cm air	B1
	<i>beta</i> electron	C1
	<i>gamma</i> electromagnetic (wave / particle)	A1
	<i>gamma</i> no charge	C1
	<i>gamma</i> not stopped (easily) or stopped by a few cm of lead or stopped a few metres of concrete	A1
10(b)(i)	(nuclei / atoms with) same number of protons (in nucleus)	B1
	different number of neutrons	B1
10(b)(ii)	ionisation or can penetrate (through air / boxes) or has high energy	B1
10(b)(iii)	stopped by box / (air-tight) bag, instruments or stopped by air	B1
10(b)(iv)	decays too quickly or has to be replaced often	B1
10(c)(i)	(alpha) repelled by nucleus	B1
	alpha and nucleus are positive	B1
10(c)(ii)	most of an atom is empty space / nucleus is only a small part of atom	B1
	alpha particle rarely passes close to nucleus	B1

13.

Question	Answer	Marks
6(a)(i)	16 protons (in nucleus) and 16 electrons in shells or 16 protons (in nucleus) 16 electrons around / outside / surrounding / orbiting nucleus	B1
	15 neutrons (in nucleus)	B1
6(a)(ii)	one more neutron (and same number of protons in nucleus) or 15 neutrons in P-31 and 16 in P-32	B1
6(b)(i)	one more proton and one neutron fewer c.a.o.	B1
6(b)(ii)	point at / line through (2.0, 1.6×10^{11})	B1
	point at (8.0, 3.0×10^{11}) or any other obvious correct point	B1
	curve of positive and decreasing gradient and through their plotted points to at least 6.0 weeks	B1
6(c)	any two from: lead shielding (e.g. lead apron / gloves / container / (lead) glass screen / (lead glass) goggles / (lead glass) glasses large distance (e.g. tongs / forceps / tweezers / manipulator / carry at centre of large box) minimum time of exposure or wear film badge	B2

14.

Question	Answer	Marks
7(a)(i)	T and Q or U and R	B1
7(a)(ii)	different number of neutrons (in nucleus)	B1
7(b)(i)	time when decay occurs / particle emitted is not known / not constant or direction in which particle is emitted is not known / can be at any angle / direction	B1

7(b)(ii)	$^{218}_{84}\text{Po}$	B1
	$^4_2\alpha$ He or	B1
7(c)(i)	electron or beta particle	B1
7(c)(ii)	(R gains or S has) one more proton	B1
	(R loses or S has) one less neutron	B1

15.

Question	Answer	Marks
7(a)	emission of (at least one of) alpha, beta and gamma (particles / rays)	B1
	from the nucleus or at random	B1
7(b)	carbon-12 6, 6	B1
	carbon-14 6, 8, 6	B1
7(c)(i)	<u>beta</u> (particles)	B1
7(c)(ii)	(does not emit) gamma	B1
	gamma passes through thick container / 6 mm (aluminium)	B1

16.

Question	Answer	Marks
10(a)	similarity: (same) number of protons	B1
	difference: (different) number of neutrons	B1
10(b)(i)	(nucleon number =) $222 - 4$ or 218 or $218 - 84$	C1
	134	A1
10(b)(ii)	(proton number =) $84 + 2$	C1
	86	A1
10(b)(iii)	it / helium atom has two electrons (in orbit around the nucleus)	B1
10(c)(i)	2.1×10^{10} (alpha-particles)	B1
10(c)(ii)	calculated value of: 2.8×10^{10} – candidate's 10(c)(i) (atoms remain) (expected value: 7.0×10^9)	B1
10(c)(iii)	$7.0 \times 10^9 / 2.8 \times 10^{10}$ or 1 / 4	C1
	2 (half-lives) or 7.6 / 2	C1
	3.8 days	A1
10(d)	alpha-particles are (very) weakly penetrating	B1
10(d)	strongly ionising	B1
	(ionisation causes cell) mutations / cancers	B1

17.

Question	Answer	Marks
7(a)(i)	(atomic particle) Q	B1
7(a)(ii)	$+1.6 \times 10^{-19} \text{ C}$	B1
7(b)	(atomic particle) Q and (atomic particle) S	B1
7(c)(i)	(atomic particle) R	B1
7(c)(ii)	(atomic particle) S	B1
7(c)(iii)	beta-(particle) emission	B1
7(c)(iv)	99 / 33 or 3 half-lives	C1
	1 / 2 ³ or 1 / 8 (remain)	C1
	7 / 8	A1

18.

Question	Answer	Marks
11(a)(i)	gravity or gravitational attraction	B1
11(a)(ii)	Any 2 from: - high pressure (very) high temperature - high particle density	B1 B1
11(b)(i)	proton and 92	B1
	neutron and 143	B1
11(b)(ii)	same number of protons	B1
11(b)(iii)	neutron hits / absorbed by uranium / nucleus	B1
	<u>nucleus</u> splits or smaller <u>nuclei</u> formed and emits more neutrons	B1
11(b)(iv)	fusion – coming together and fission splitting up or fission involves large particles / nuclei and fusion small nuclei or in fusion nuclear waste / products are less radioactive / dangerous or no / few neutrons produced in fusion	B1
11(b)(v)	90 proton number of Th	B1
	231 and 4 nucleon numbers	B1
11(c)(i)	P marked where beam of alpha-particles hits screen	B1
11(c)(ii)	fewer flashes (as one moves from centre) or very few / only some flashes on same side as source / on left	B1
11(c)(iii)	Any 2 bullet points from <u>nucleus</u> is small or atom (mostly empty) space - most mass of atom in nucleus or mass is concentrated in small point / region or nucleus is dense or nucleus is massive - charge concentrated in small region or (positive) charge found in nucleus or nucleus has all the (positive) charge of the atom	B1 B1

19.

Question	Answer		Marks
11(a)(i)	Geiger Muller (tube / counter / detector)		B1
11(a)(ii)	keep distance (away from source) or use absorber or use for short time		B1
11(b)(i)	time between emissions varies / unpredictable or different readings (when repeated for same time)		B1
11(b)(ii)	repeated count for a specified time	measure time when count / event occurs	B1
	result not the same	not always the same	B1
11(c)(i)	radiation when there is no source or naturally occurring (radiation) or always present		B1
11(c)(ii)	<u>measure</u> count (rate) with <u>no source</u>		B1
	subtract from (measured) count (rate for same time)		B1
11(d)(i)	4.5 cm		B1
11(d)(ii)	alpha		M1
	range of alpha less than 5–7 cm (in air)		A1
11(d)(iii)	air atoms / molecules lose / gain electrons (to become ions)		B1
11(e)	points plotted at (0, 300) and (40, 150)		B1
	points plotted at (80, 75) and (120, 37.5)		B1
	smooth curve joining at least 3 correct points		B1

20.

Question	Answer	Marks
6(a)	32 counts / minute	B1
6(b)(i)	second (beta-particles) and third (gamma-rays) boxes ticked	B1
6(b)(ii)	1 28	B1
	2 32	B1
6(c)(i)	(average) time (taken for)	M1
	count rate / number of nuclei / number of atoms to halve	A1
6(c)(ii)	some readings are bigger than those before / readings fluctuate	B1
	half-life / 5.3 years too long or 5 / 6 minutes too short	B1

21.

Question	Answer	Marks
10(a)(i)	same number of protons or same number of electrons	B1
10(a)(ii)	different number of neutrons	B1
10(b)(i)	${}^4_2\alpha$	B1
	${}^{238}_{92}\text{U}$	B1
	${}^{235}_{92}\text{U}$	B1
10(b)(ii)	(no. of half-lives =) $1.2 \times 10^5 / 2.4 \times 10^4$ or 5 (half-lives)	C1
	relevant halving or $1/2^5$ or $1/32$ or 6400 / 32	C1
	200 counts / second	A1
10(b)(iii)	1 curving (only) downwards with correct curvature	M1
	no straight section (by eye) anywhere in magnetic field	A1
	2 (Fleming's) left-hand rule or motor rule or rule described or moving charge is a current	B1
10(c)(i)	splitting of a <u>nucleus</u>	B1
10(c)(ii)	neutron fired at a plutonium nucleus or plutonium nucleus absorbs neutron	B1
10(c)(iii)	1 no CO_2 / SO_2 / nitrogen oxides / greenhouse gases emitted or do not contribute to global warming / acid rain	B1
	2 reliable / not intermittent or smaller land area	B1

22.

Question	Answer	Marks
11(a)(i)	detector (need not be named) and absorber or ruler (to measure distance in air) OR cloud chamber (need not be named) and source inside	B1
	suitable detector named or labelled, e.g. GM detector, Geiger counter, cloud chamber, photographic film/paper, spark counter or counter / ratemeter connected to labelled detector	B1
11(a)(ii)	count with source (alone) or listen to clicks / sound	B1
	count with source or listen <u>to clicks</u> and suitable, named absorber or apply magnetic / electric field at right angles to beam	B1
11(a)(iii)	count decreases when absorber used or no / background radiation detected or particles found to be deflected in correct direction for alpha	B1
	(only) alpha-particles absorbed / stopped (by air / paper) or deflection is correct for alpha-particles	B1
11(b)	ionisation (in air) alpha (>) beta (>) gamma	B1
	penetration gamma (>) beta (>) alpha	B1
11(c)(i)	(detector) works for longer or does not need replacing (as often)	B1
11(c)(ii)	any halving seen	C1
	3 half-lives or three halvings seen	C1
	1290 or 1300 years	A1
11(d)(i)	kills / damages cells or causes cancer or causes mutations	B1
11(d)(ii)	ANY 2 from (source) further away (from the body) - less radiation / fewer particles pass through / reach body / hand - alpha particles stopped / reduced by air - particles spread out from source	B2

23.

Question	Answer	Marks
8(a)(i)	2 protons and 2 neutrons (joined together)	B1
8(a)(ii)	it is (positively) charged	B1
	it pulls electrons from molecules of air or knocks electrons from the molecules of air or gains electrons from air molecules	B1
8(a)(iii)	1 stronger and 2 stronger	B1
8(b)(i)	idea of halving	C1
	($N = $) $4.8 \times 10^7 / 1.5 \times 10^6$ or 1 / 32 or 5 (half-lives)	C1
	$1.6 / 1.65 / 1.7 \times 10^6$ s	A1
8(b)(ii)	radioactive emission is a random process	B1

24.

Question	Answer	Marks
11(a)(i)	rocks / soil / Earth's surface / building materials / radon (gas) / waste from a nuclear power station / weapons testing	B1
11(a)(ii)	cancer / mutation / cell damage / gene damage or adds to / affects experimental readings / count rate or causes ionisation	B1
11(a)(iii)	protons 2	B1
	neutrons 2	B1
11(a)(iv)	alpha-particles absorbed / stopped by / cannot penetrate air / atmosphere or scattered (by air) or cause ionisation (and slow down)	B1
11(a)(v)	continuous curve deflected either clockwise or anticlockwise within shaded area	B1
	arrow or other indication to show anticlockwise deviation within shaded area	B1
11(b)(i)	time taken to halve	C1
	time taken for the activity / count (rate) / number of atoms / number of nuclei to halve	A1
11(b)(ii)	any halving seen, e.g. 200–100 or 3 half lives	B1
	17 100 years	B1
11(b)(iii)	too little carbon-14 left or all decayed or shows large reduction	B1
11(c)(i)	different number of neutrons or different mass or different nucleon number	C1
	carbon-14 has two more neutrons	A1
11(c)(ii)	same number of protons	B1

25.

Question	Answer	Marks
8(a)	(high frequency) electromagnetic wave / ray	B1
8(b)	top line of beta 0	B1
	Xe proton and neutron numbers both balance	B1
8(c)	gamma rays pass out of / are not stopped by body	B1
	less chance of cell or gene damage / cancer / radiation sickness / mutation / ionisation (of body tissues)	B1

26.

- (a) 94 electrons **and** 94 protons B1
 144 neutrons B1
 (only) electrons in orbit / surrounding nucleus **or** (only) protons and neutrons in nucleus B1
- (b) (i) (beta-particles) weak(er) B1
 (beta-particles) strong(er) B1
- (ii) any **two** lines from
 glasses / goggles **or** lead container / shield / clothing / gloves
 tweezers / manipulator / carry in large cardboard box
 minimise time of exposure / film badge B2

27.

- | | | |
|---------|---|----|
| (a) (i) | 11 protons and 11 electrons | B1 |
| | 13 neutrons | B1 |
| | electrons in orbit/surrounding nucleus or neutrons and protons in nucleus | B1 |
| (ii) | one more neutron (in sodium-24) or one fewer neutron in sodium-23 | B1 |
| | | |
| (b) (i) | electron | B1 |
| (ii) | ${}^0_{-1}(\beta)$ cao | B1 |
| | ${}^{24}_{12}(\text{Mg})$ | B1 |
| | ${}^{24}_{12}(\text{Mg})$ | B1 |
| | | |
| (c) | electromagnetic (radiation / rays / waves) | M1 |
| | (very) high frequency / energy or (very) short wavelength | A1 |
| | | |
| (d) (i) | path curving upwards | B1 |
| (ii) | path continues in straight line | B1 |
| (iii) | beta-particle charged or gamma-ray uncharged | B1 |
| | | |
| (e) | long enough or short enough | B1 |
| | to take measurements so the body is not irradiated for long | B1 |

28.

(a) (nucleus / nuclide / atom) with same number of protons

(b) (i) 2

(ii) neutron

(iii) 2

(iv) 4

(c) nuclei repel

or like / positive charges repel

(needs) high kinetic energy / speed (to overcome repulsion)

(d) ANY 3 lines from

(dust / gas) collapses / comes together / clusters / condenses

gravitational attraction or gravity mentioned

B1

temperature rises or KE (dust / gas) increases

B1

(nuclear) fusion occurs

B1

equilibrium established as radiation pressure / outward

force balances inward force

B1

(e) (i) time for a quantity to halve

time for (radio)activity / count rate / number of atoms / number of nuclei to halve

(ii) any relevant halving seen, e.g. 16 000 / 2
1000

29.

(a) 127 n(eutrons) **and** 82 p(rotons) B1
 82 e(lectrons) B1
 electrons in orbit around nucleus/in shells around nucleus **or** around
 neutrons and protons **or** neutrons and protons in nucleus B1

(b) (i) more protons **and** fewer neutrons **or** one more proton C1
 one more proton **and** one fewer neutron **or** neutron becomes proton
 (and electron/beta-particle) A1

(ii)

		
gamma / γ	beta / β	alpha / α

reversed order 1 / 2

correct order 2 / 2 B2

(iii) (circular by eye) curve from beginning of field **not** with straight line
 initially B1
 upward curve (in field) B1

(c) (i) (radiation) that is always present **or** occurs everywhere **or** cannot be
 eliminated **or** from environment / surroundings / natural sources / air B1

(ii) two separate sources: rocks (e.g. radon / ground), outer space
 (e.g. cosmic rays), man-made sources (e.g. nuclear waste / fall-out) B2

(iii) 2 half-lives (implied) **or** $\frac{1}{4}$ seen **or** 76 (counts / minute) C1
 19 **or** 23 (counts / minute) C1
 35 counts / minute A1

30.		
(a)	(i) (atoms) 3 and 4	B1
	(ii) (atoms) 3 and 5	B1
	(iii) (atoms) 3 and 4	B1
(b)	(i) 17	B1
	(ii) 35	B1
(c)	(i) two separate sources: rocks (e.g. radon), outer space (e.g. cosmic rays), man-made sources (e.g. nuclear waste/fallout)	B2
	(ii) 22 counts/minute	B1
	(iii) 27 counts/minute	B1
	(iv) use of 27/2 or 27/4 or 27/8 from 85 to 90 days (inclusive)	B1 B1
(d)	(i) (background count-rate is) reduced not to zero or not stopped or (some) gamma-rays in background count	B1 B1
	(ii) not sensible all the beta-radiation would be absorbed or no beta-radiation reaches the detector	M1 A1

31.

- (a) (i) 51 B1
- (ii) more protons than electrons B1
or different number of protons and electrons
positive and negative do not cancel B1
- (iii) 25 protons B1
a different number of neutrons B1
- (b) (i) 147 B1
- (ii) α has mass number 4 B1
 α has proton number 2 B1
correct proton number for U ecf their value for α B1
- (c) (i) alpha particles only travel a short distance in air B1
or alpha particles stopped / scattered / deflected by air
or alpha particles ionise air
- (ii) particles come off in different directions B1
or not emitted in one line / as a ray
or not all the particles pass through the slit
- (iii) B correct shape and deflected more than A B1
- (iv) particles **close to / fired at** the nucleus are deflected (back) / repelled B1
some particles pass (straight) through B1
a few particles come back / large deflection or most pass (straight) through
(with little deviation) B1
and how this explains the nucleus is small

32.

- (a) same element **or** same number of protons / atomic number B1
different / particular number of neutrons **or** nucleons B1

- (b) (i) 38 **cao** B1

- (ii) 52 **cao** B1

- (c) $^{90}_{39}\text{(Y)}$ **or** $^{90}\text{(Y)}$ and $^0_{-1}\text{(}\beta\text{)}$ B1

- $^0_{-1}\text{(}\beta\text{)}$ $^{90}_{39}\text{(Y)}$ and $^0_{-1}\text{(}\beta\text{)}$ B1

- (d) (i) 87/29 **or** 3 (half-lives) **or** $6.0 \times 10^8/8$ C1
 7.5×10^7 A1

- (ii) any detector B1
corresponding detection method B1

detector	film	(solid-state) detector	GM-tube	ionisation chamber	scintillation counter	cloud chamber
detection	fogged	count/reading	count/reading	count/reading	count/reading	track seen

- no reduction with paper **or** (use of) electric/magnetic field **or** describe pattern of track M1

- complete reduction with aluminium/lead **or** correct deflection of track in electric/magnetic field **or** no other track A1

- (iii) 1. unpredictable **or** not be known in advance **or** no set time between emissions **or** fluctuates **or** not fixed **or** counts obtained varies B1

2. any **two** from:
direction / in space
time
which nucleus decays

B2

- 33.**
- (a) neutrons and protons together and alone in the middle B1
 5 protons B1
 7 neutrons (if protons and neutrons unlabelled 1/2) B1
 5 electrons **and** electrons surrounding nucleus B1
- (b) (i) 6 B1
 (ii) 12 B1
- 34.**
- (a) (i) electron(s) and proton(s) [B1]
 (ii) neutron(s) and proton(s) [B1]
- (b) (i) top box 14 [B1]
 bottom box 7 [B1]
- (c) (i) sensible halving seen, e.g. 2.4 → 1.2 **or** two halves clear **or** $\frac{1}{2} \times \frac{1}{2}$ seen [C1]
 11 400 **or** 11 000 years [A1]
- 35.**
- (a) (i) 53 protons **not** if also 53 electrons in nucleus [B1]
 78 neutrons **or** 131 protons and neutrons [B1]
- (ii) emission of at least one of alpha particle, beta particle or gamma ray [B1]
 emission from the nucleus **or** breakdown of nucleus [B1]
- (b) (i) random emission indicated [B1]
- (ii) average 2772 **or** 2773 **or** 2770 **or** 2800 **or** 1/8 used **or** 3 clear halvings seen [C1]
or $(\frac{1}{2})^3$ seen **not** halving of 131 or 53 [A1]
 value between 330 and 360

36.

- (a) (i) any two of:
 minimise time of exposure
 lead clothing (e.g. lead gloves **not** radioactive suit)
 tongs, manipulator, forceps, tweezers
 behind protective/lead glass/shield
 wear film badge B2
- (ii) (radioactive emission) random/unpredictable (process)
 (e.g. background radiation is random; **ignore** spontaneous) B1
- (b) penetration strong(er) **or** penetrates casing (accept α **or** β **or** both;
ignore larger range) B1
 (more) weakly/slowly ionising B1
 either explained: harms health **or** hazardous **or** dangerous
or air is not ionised **or** sounds all the time (accept doesn't work) B1

37.

- | (a) | Penetration | Magnetic/electric field | Cloud chamber | Spark counter | |
|-----|---|---|--|--|----|
| | diagram:
sample,
detector, small
gap | diagram:
sample, detector,
magnet | diagram:
sample, cloud
chamber | diagram:
sample, spark
counter, small
gap
labelled or clear | B1 |
| | (insert/remove)
(a sheet of)
paper/card/Al
foil (in gap) | (insert/remove)
magnet | sample in
cloud
chamber | sample near
to counter | B1 |
| | no change in
count | increased count in
correct direction | no short,
straight,
dense tracks | no sparks | B1 |
- (b) any **two** of:
 minimise time of exposure
 lead clothing (e.g. lead gloves **not** radioactive suit)
 forceps, tweezers, tongs, manipulator
 behind protective glass/shield
 wear film badge B2

- 38.
- (a) (i) 6 protons B1
 8 neutrons B1
 6 electrons outside nucleus **or** 6 electrons and protons & neutrons inside nucleus B1
- (ii) different number of neutrons B1
 same number of protons (ignore electrons) B1
- (b) (i) 2 half lives seen e.g. $8 \rightarrow 4 \rightarrow 2$ C1
 number of carbon atoms 2×10^{20} A1
 number of nitrogen atoms 6×10^{20} **or** $(8 \times 10^{20} - N_C)$ atoms B1
- (ii) many half lives **or** has decayed (too much) **or** very few atoms (of C) left B1
- (c) (i) background count/rate taken without source B1
 any count taken over any measured time e.g. 1 minute
or any rate determined (**allow** read ratemeter) B1
 take count/rate with aluminium **between source and detector** B1
 at 5 mm count/rate goes to background/constant/zero **when** corrected for background B1
- (ii) for protection (of the class/teacher e.g. to avoid cancer) **or** gamma-rays cannot pass through **or** to stop particles (hitting class/teacher) (**ign.** just "for safety") B1
- (iii) gamma-rays not stopped by/pass through (5–10 mm) aluminium B1

- 39.
- (a) solid-state detector/GM tube/ionisation chamber/scintillation counter/spark counter/spinthaloscope B1
 count **or** count-rate **or** reading referred to B1
 (some) detection with appropriate blocking in the way **or** same reading/track in electric/magnetic field B1
OR
 film B1
 develop B1
 (some) detection with appropriate blocking in the way **or** same reading/track in electric/magnetic field B1
OR
 (diffusion) cloud chamber B1
 track **seen/looked for/formed** B1
 pattern of track described B1
- (b) any **two** lines:
- one **distance** method: tongs/robotic arm/carry in large box
- one **protection** method: lead suit/lead gloves/lead boxes/shield
- one **time** method: reduced time/wear badge B2
- 40.
- (a) (i) two protons
 OR has charge +2(e)
 OR helium nucleus OR He nucleus B1
- (and) two neutrons
 OR has mass 4 (u)
 OR symbol ${}^4_2\text{He}$ B1
- (ii) electromagnetic (particle/wave) M1
 high frequency/high energy/low wavelength A1
- (b) (i) 1. alpha identified (as the reason) B1
- (alpha) particles **stopped/blocked/absorbed** (few cm air)
 OR distance covered by/range of (alpha) particles (in air) is small/a few cm B1
2. experiment takes time in some way
 OR otherwise count falls (during half life) B1

- (ii) keep distance (e.g. use forceps/tongs, do not point at person/eyes)
OR use absorber (e.g. lead covering)
OR place in store when not in use; use for short time
OR wear badge B1
- (c) (i) YES (alpha particles present)
and count falls **with paper** in some way B1
- (ii) NO (beta particles) M1
when (5 mm) Al used **and**
no **further/more/extra** reduction OR no difference A1
- (iii) YES (gammas present)
and gammas pass through (5 mm) Al **or** 820 after Al B1
- (d) ANY 2 lines
cosmic rays; the Sun; outer space
rocks (e.g. granite); stones; soil; buildings; food
radon/thoron/carbon-14 (gas)
weapons tests; nuclear bombs
leaks from (nuclear) power stations
nuclear waste B2
- (e) cancer (accept any specific cancer); tumours
radiation sickness; burns; mutations;
genetic problems; damage to DNA/chromosomes
cell damage (e.g. kills cells, cures cancer); birth defects
sterility; hair loss B1

41.

- (a) (gamma) produces little or no ionisation; passes out of detector; requires shielding; reaches people B1
- (b) (i) at least 3 lines between plates and in middle at least one **straight, vertical** line B1
 correct curvature at edges B1
 at least one arrow down and no arrows wrong B1
- (ii) alpha charged (positively); alpha repelled by positive/attracted by negative B1
 deflected down/towards positive (plate) B1
- (c) (gamma) undeviated; straight line B1
 (gamma) uncharged B1
- (d) (i) any attempt at halving or 3 **half-lives** seen C1
 $1/8$; 0.125; 12.5% A1
- (ii) decays too fast; have to replace source often; current falls too quickly; detector only works for a short time B1
- (iii) 1. any TWO of:
 number of protons
 number of electrons
 charge on nucleus B2
2. number of neutrons B1
 nucleon or mass number or mass B1
 (americium-242 has **one** extra neutron gets 2 marks)

42.

(a) (nuclear) fission B1

(b) (i) 1...143 B1

2...36 B1

3...141 B1

(ii) $(E =) mc^2$ C1

$3.1 \times 10^{-28} \times (3.0 \times 10^8)^2$ or $3.1 \times 10^{-28} \times 3.0 \times 10^8$ and $(E =) mc^2$ C1

$2.8(2.79) \times 10^{-11} \text{ J}$ A1

(c) any five of:



(one mark for three correct boxes)

(splitting produces) kinetic energy of neutrons

further splitting/chain reaction

energy/heat produced/from reactor/reaction or from neutrons

coolant gets hot

energy to boiler/water or water heated or heat in water implied

water boiled or steam produced B5

(d) (i) time for something to halve C1

time for (radio)activity/count rate/number of atoms/nuclei to halve A1

(ii) one appropriate precaution:

short exposure time

safety/protective suit/gloves/clothes or lead boxes

large distance/(long handled) tool/forceps/tongs

robotic/mechanical handling

film badge B1

- 43.
- | | | | | |
|-----|------------------------|----|---|----|
| (a) | $^{131}_{54}\text{Xe}$ | OR | ^{131}Xe and $^0_0\beta$ | B1 |
| | $^0_{-1}\beta$ | | $^{131}_{54}\text{Xe}$ and $^0_{-1}\beta$ | B1 |
- (b) (i) downward curve B1
- (ii) horizontal line B1
- (c) any **two** of:
 direction/space (of emission)
 time/frequency (of emission) **or** period/interval between emissions **or** different counts (in same time)
 nucleus that decays is unpredictable B2
- 44.
- (a) fission cao B1
- (b) neutron hits/goes inside (U) **nucleus** B1
 atom/nucleus/particle/uranium/nuclide splits/forms daughter nuclei **and** emits neutrons/energy B1
- (c) (i) emits particles // emits **ionising/nuclear** radiation // spontaneous or random emission (of radiation) // **atom/nucleus** decays B1
- (ii) long time to decay // radioactive for a long time // decays slowly B1
 long time for any quantity to halve
 halving of:
 count, count rate, emissions, (number of) nuclei, (number of) atoms, activity B1

45.		
(a)	(i) 24	B1
	(ii) charges (of electrons and protons) cancel // protons and electrons have the same (size of) charge but opposite in sign	B1
(b)	(i) nuclear at start // nucleus loses energy to kinetic/e.m. energy (condone light/photon/ γ)	B1 B1
	(ii) beta 0 at top	B1
	beta -1 at bottom	B1
	Mg 24 at top	B1
	Mg 12 at bottom	B1
(c)	(i) 600 at 15 hours	B1
	300 at 30 hours	B1
	line of decreasing gradient (not if it cuts time axis before $t = 30$)	B1
	(ii) emission is random // not predictable // not regular // exptl. error	B1
	(iii) 4800 (counts per minute) // 80 counts /sec	B1
	(iv) GM tube/solid state detector/cloud chamber	M1
	ratemeter/data logger // counter/scalar and stopwatch/timer	A1

46.		
(a) (i)	15	B1
(ii)	32	B1
(b) (i)	$^{32}_{15}\text{P} (\rightarrow)$	B1
	superscripts: 32 on S and 0 on beta (allow e)	B1
	subscripts: 16 on S and -1 on beta (allow e) (just $^{32}_{16}\text{S } 1/2$)	B1
(ii)	electron	M1
	high speed or from nucleus or causes ionisation	A1
(iii)	record/measure background reading/count/radiation	B1
	sample near named detector	B1
	interpose paper/card/less than 5 cm air and no change in reading	B1
	interpose 2 mm – 20 mm of aluminium and reading = background	
	n.b. points may be made on a diagram, other methods marked analogously	B1
(c) (i)	time for some measurable quantity to halve	M1
	number of atoms/no. of nuclei/activity/count rate	A1
(ii)	$350 \div 1400$ or $\frac{1}{4}$ or 2 (half-lives)	C1
	28.6 days	A1
47.		
(a) (i)	Geiger Muller/GM tube or any other gamma detector	B1
(ii)	keep distance e.g. forceps/tongs	
	barrier accept gloves, lead suit, metal container	
	time of use accept use badge ANY ONE	B1
(iii)	(with source) take a count for a time	
	or take count rate (from a ratemeter/meter/count meter/counter)	
	or read ratemeter (connected to GM tube) or record number of tracks (chamber)	
	or count clicks in a time or note time when click occurs	B1
	or several readings taken or readings fluctuate ignore readings random	
	or time when click occurs varies	B1
(b)	electromagnetic	B1
	(wave/ray/particle) high frequency or small wavelength	B1

- 48.
- (a) electron B1
 negative B1
 electromagnetic (high frequency wave/particle/photon) B1
 neutral/none B1
- (b) (i) time taken to halve M1
 activity **or** number of atoms/nuclei **or** count (rate)
 (**ignore** radioactivity/mass/volume/amount/number of particles/molecules/
 a nucleus to halve) A1
- (ii) alpha stopped by body/flesh/skin **or** cannot penetrate body/skin **or** causes
 damage to body (1 max for damage) B1
 gamma penetrates body/not absorbed **or** can be detected outside body **or**
 causes less/no damage to body (1 max for damage) B1
- (iii) takes time for isotope to spread/investigation/experiment (so 6 min too short) B1
- (iv) radioactive for longer/more dangerous/more damage/causes damage B1
- (c) (i) (radioactive emission is) random B1
- (ii) 3200 seen (as average) C1
 attempt to halve e.g. 3202 → 1601 **or** 4 half-lives C1
 52 hours A1
- (iii) rocks/cosmic rays/radon gas/nuclear fall out B1
- 49.
- (a) (i) 3 **cao** B1
- (ii) 208 **cao** B1
- (iii) 11 **cao** B1
- (b) (i) 17 **cao** B1
- (ii) 20 **cao** B1

50.		
(a)	number of protons and neutrons	B1
	protons and neutrons in the nucleus	B1
(b) (i)	2	B1
(ii)	4	B1
(iii)	90 or 92– (i) and (iv) 234 or 238– (ii)	B1
51.		
(a)	fusion	B1
(b) (i)	mass decreases or product/nuclei/atoms less massive	B1
	mass becomes/converted to energy	B1
(ii)	$E = mc^2$	B1
	$6.6 \times 10^{-29} \times (3.0 \times 10^8)^2$	C1
	5.9×10^{-12} or 5.94×10^{-12} J	A1
52.		
(a)	different number of neutrons/ mass number	B1
(b)	time taken to halve	M1
	(number of) nuclei/atoms/activity/count (rate)	
	(not radioactivity/amount/mass/substance/clearly one nucleus/particles)	A1
(c)	mark on graph at 2000 or at two suitable points	B1
	12 s	B1
53.		
(a)	emission of at least one of alpha, beta, gamma (particles)	M1
	from the nucleus or at random	A1
(b) (i)	background stated or explained	B1
(ii)	not radioactive	B1
	average the same or 93 total on both sides	
	or two increase and one decreases or variation explained	B1